

A Codebook

1. Motivation

1.1. Contribution of the Paper

- User study
- Literature review
- Scoping review
- Qualitative studies

1.2. Motivation of the Paper

- Improve survey data quality
- Impacts of fraud on data quality
- Research is needed because of excessive fraud in surveys
- Characterize fraudulent behaviors
- Propose a detection or mitigation strategy for survey fraud
- Study the impact of LLMs or AI-mediated fraud

2. Conceptualization of Fraud

2.1. Does the paper define *fraud*?

- Definition of survey fraud
- Definition of careless response
- Traditional bot
- AI-based bot
- Location-based
- Multiple submissions
- Inattentive

2.2. Fraud Vector (Type of Fraudulent Behavior)

- **A. Eligibility / Identity Fraud**
 - Misreported demographics
 - Age misrepresentation
 - Location spoofing (e.g., VPN)
 - Identity spoofing / false claims
 - Multi-accounting / duplicate submissions
- **B. Response Integrity Fraud**
 - Careless responding
 - Mischievous responders
 - Bot-generated

3. Detection & Mitigation Strategies

3.1. Stage of Detection

- Recruitment phase
- In-survey phase
- Post-hoc (after data collection)

3.2. Techniques Used

- **A. Metadata-based Methods**
 - IP address / geolocation
 - VPN / proxy detection
 - Device / browser fingerprinting

- Time
- Server-side logs

- **B. Response-based Methods**

- Attention checks, instructional manipulation checks
- Error-count checks
- Randomness filtering
- Semantic similarity / embedding analysis
- LLM detectors / AI-content detection
- Stylometry / writing-style analysis / internal consistency
- Honesty checks
- Low-incidence questions
- Knowledge questions

- **C. Platform-Level Mechanisms**

- Quality controls, platform qualification filters
- Reputation scores / approval ratings
- CAPTCHA / ReCAPTCHA

- **D. Manual Review**

3.3. Detection Response

- Prevent response completion
- Remove data
- Deny compensation
- Track participant
 - Internally
 - Report to platform

4. Output, Replicability, and Impact

4.1. Primary Output Type

- Tool or software system
- New algorithm or detection method
- Empirical measurement study
- Guidelines or recommendations
- Suggested questions, examples of questions

4.2. Reproducibility / Artifact Availability / Supplemental Material

- Code provided
- Dataset provided

5. Examples of Impacts of Survey Fraud